

Adaptive Trial Design for the RAS-Inhibitor Era

Combinations, Resistance, and Biomarker-Guided Therapy in GI Oncology

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Today's argument

New RAS therapies need new kinds of trials.

Methods to optimize combination doses are established. Designs that track resistance and adapt the trial in response are enrolling.

What's missing is making those designs practical to calibrate and run.

The argument of the talk. The science of RAS-era combinations puts real demands on trial design — combination dose-finding under non-monotonic dose-response, and resistance tracking with in-trial adaptation. Modern Bayesian adaptive designs answer both, at different maturity levels: dose-optimization methods are established, resistance-tracking designs are enrolling. What's been missing is the infrastructure to make them practical, and that's where BATON closes the gap. BATON is the closing tool, not the headline. Note the deliberate asymmetry in the fragment — “established” vs “enrolling” — this prevents the resistance-side overclaim that would land badly with a SPORE audience who knows EVOLVE hasn't yet reported.

The clinical case

Why combinations, and why now?

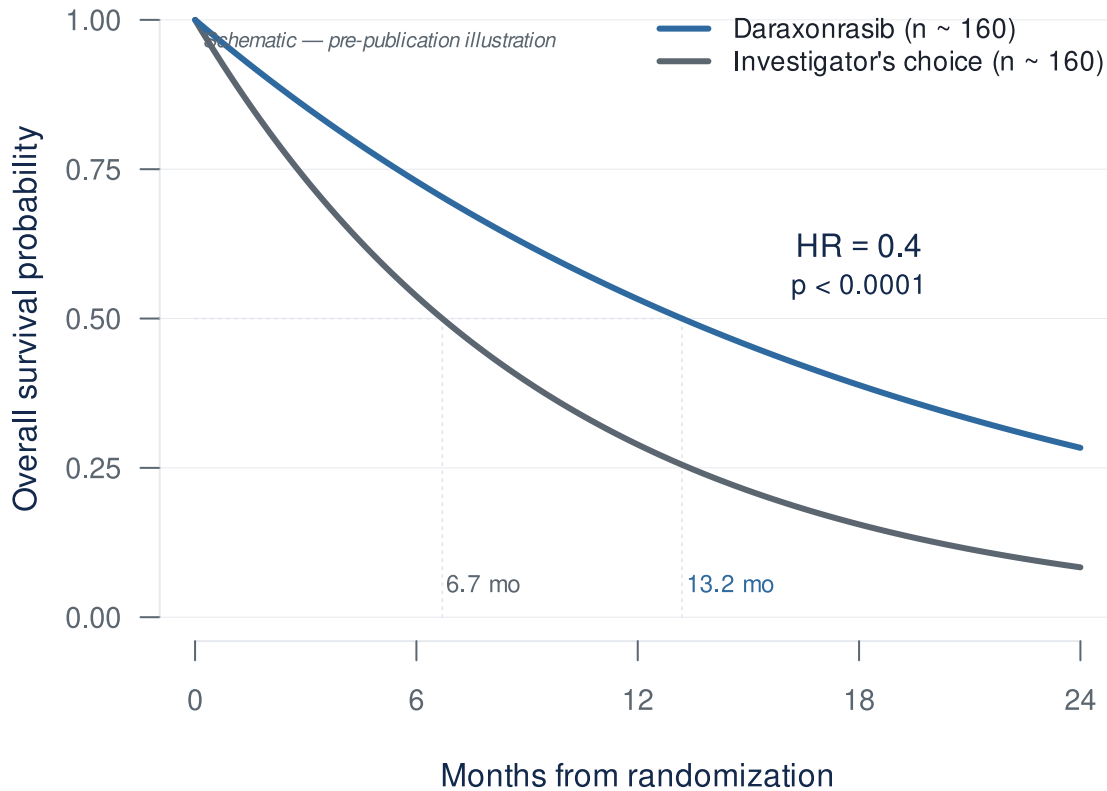
Three slides on the clinical pressure before we get to design problems.

The RAS revolution has arrived for PDAC

- Daraxonrasib (RMC-6236): oral, multi-selective RAS(ON) inhibitor
- **RASolute 302** (2L mPDAC)¹:
 - **Median OS: 13.2 vs 6.7 months**
 - **HR 0.40 — 60% reduction in risk of death** ($p < 0.0001$)

- ASCO 2026 Plenary, May 31 (Wolpin); *NEJM* same day

O'Reilly et al., *NEJM* 2026 (RASolute 302) · ASCO Plenary Abstr LBA5



For pancreatic cancer this is the largest survival signal in a generation. Final analysis was presented at the ASCO Plenary on May 31 and published in *NEJM* the same day. Cite the *NEJM* paper, not a press release — this room will care about that. Two numbers to have in your pocket for Q&A. Median follow-up was 8.5 months at the February 10 data cutoff, so if anyone presses on maturity that's your number. And the 13.2 vs 6.7 figure is overall ITT; the dual primary endpoint was OS and PFS in the RAS G12-mutant subgroup (228 vs 231 patients), where HR was also 0.40 ($P = 5.9e-10$). If a discussant draws the ITT-vs-primary distinction, the answer is that the 0.40 benefit held in both — robustness across RAS mutation status is the whole point of a multi-selective RAS(ON) inhibitor. That robustness is a strength to lean on, not a vulnerability. Whatever else you take from this talk, this result will reshape the GI oncology trial landscape over the next two to three years — every program in this room will soon be deciding what to combine with this drug, in which biomarker-defined patients.

...but monotherapy has visible ceilings

RMC-6236-001 (NEJM 2026)²

Full cohort (n=168 PDAC, doses 300 mg):

Metric	Value
Grade 3 TRAEs	30%

At 300 mg (the Phase 3 monotherapy dose):

Metric	Value
Grade 3 TRAEs	~ 34%
Required dose modification	48%
TRAE-related discontinuation	0%

48% is the headroom number. Every combination partner you add is dose-modification room you no longer have.

Wolpin et al., *NEJM* 2026 (RMC-6236-001 PDAC cohort)

Three numbers worth keeping in mind. Across the full cohort of 168 patients on doses up to 300 mg, about a third have Grade 3 or worse toxicity. No patient discontinued because of drug toxicity — that’s genuinely remarkable for an active drug in this disease. The qualifier matters; everyone with metastatic PDAC eventually comes off for progression, but nobody had to come off this drug for tolerability. Now narrow to the 300 mg dose specifically, which is the Phase 3 monotherapy dose: Grade 3 toxicity is closer to 34%, and forty-eight percent of patients required dose modifications. That 48% is the load-bearing number for the rest of this talk. For a drug being positioned as the backbone of future combinations, that’s the headroom you start spending the moment you add a partner. The combination already moved to 200 mg — informally acknowledging the same headroom problem without searching for the actual optimum, which is the next slide. The question isn’t “is daraxonrasib tolerable as monotherapy?” — clearly it is. The question is “at what dose, on what schedule, with what partners, in which patients?” That’s a trial-design question, and the rest of this talk is about answering it.

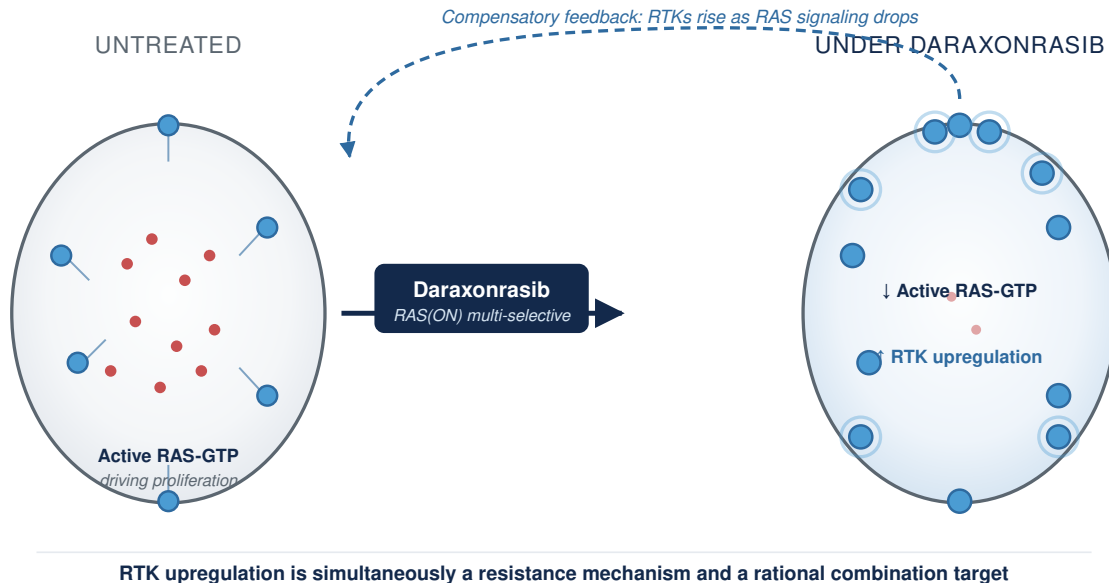
Biomarker-driven combinations are the next move

- Daraxonrasib drives **RTK upregulation** on PDAC cell surface (Zhuang/Singh/Aronchik, Revolution Medicines, AACR RAS 2026)³

- Their framing: rational, biomarker-driven combinations to *pre-empt or overcome emergent resistance*³
- **1L combination cohort** (daraxonrasib **200 mg** + GnP, n=40, AACR 2026)⁴:
 - ORR: **58%** · DCR: **90%** · 6-mo PFS: **84%** · 6-mo OS: **90%**

Dose dropped from 300 mg to 200 mg for combination — but RMC-GI-102 fixed daraxonrasib at 200 mg without searching the joint surface. Informal OBD reasoning.

Zhuang et al., *Cancer Res* 2026 (AACR RAS) · Wolpin et al., AACR 2026 LB407



Mallika Singh's group at Revolution Medicines has published the mechanistic case for what comes next. Inhibiting active RAS upregulates receptor tyrosine kinases on the cell surface — which is simultaneously a resistance mechanism and a rational combination target. Their explicit framing is that combinations should be biomarker-driven and resistance-preemptive. This is the manufacturer staking out the next decade of the program. And the first combination cohort has already delivered — 58% ORR with gemcitabine/nab-paclitaxel in first-line metastatic PDAC, 90% disease control, 84% six-month PFS. The combinations are coming. Many of them.

What 3+3 assumes, and where each breaks for combinations

Four assumptions underneath conventional dose-finding. None survive RAS combinations:

1. **More is better.** *Often it isn't.*
2. **Toxicity comes from one drug.** *Combinations don't work that way.*
3. **DLTs in Cycle 1 define safety.** *Late-onset toxicity governs long-term tolerability.*

4. **A combination behaves like the sum of its parts.** *It doesn't, on either axis. Synergy can put the optimum below both single-agent doses.*

The next section walks through what replaces each.

DELIVERY: this is a ten-second roadmap slide, not a verdict. Move through it fast. The convincing happens on the next slides where I have the examples — sotorasib + pembrolizumab, RMC-GI-102, PANGEA's OBD-below-MTD hypothesis. This slide just tells the audience the shape of the next ten minutes. Frame it as a preview: four assumptions, four places they break for combinations, each one mapped to a slide that replaces it. Assumption 1 maps to Project Optimus and the OBD paradigm. Assumption 2 maps to the sotorasib + pembrolizumab hepatotoxicity story. Assumption 3 maps to the late-onset toxicity finding on the same slide — 88% outside Cycle 1. Assumption 4 maps to PANGEA's central hypothesis, that the optimum sits below both single-agent doses. WHY THIS SCOPING: 3+3 is fine for a cytotoxic — it answered the question it was built to answer. The point is not that it is wrong, it is that the assumptions it rests on mis-specify the problem for RAS combinations. Specifically: “more is better” — the conventional heuristic that MTD is the dose to carry forward — is the one a clinical audience knows is unreliable; OBD on the next slide is what replaces it. The genuinely counterintuitive one is assumption 4. That the combination optimum can sit below both single-agent doses runs against intuition — less of each than you'd give alone — and that is the payload of this slide. The other three are scaffolding. For RAS specifically, the RTK feedback we saw two slides ago is a mechanism for non-additive interaction, so the synergy claim is biologically grounded, not just statistical caution. PANGEA's hypothesis is the local case-study version of assumption 4.

The dose-finding problem

Why MTD fails, and what replaces it

Why combinations break the rules you grew up on

- **3+3 was built for single-agent chemo**⁵. Combinations break its assumptions. Non-monotonic dose-efficacy, joint toxicity, late-onset DLTs.
- **In practice, combination dose-finding collapses to one drug.** Pick agent A's dose, fix it, add agent B. The joint efficacy-toxicity surface is **almost never searched** — in a systematic review of 162 combination trials, **88% used 3+3 and all but one used a single-agent design**^{6,7}.
- **Amgen did exactly this with sotorasib + pembrolizumab.** A Bayesian model guided escalation for the single agent⁸. For the combination they fixed the IO dose, varied only sotorasib, and never modeled the joint surface. The defining toxicity was synergistic

and mostly late: **88% of Grade 3-4 hepatotoxicity fell outside the Cycle 1 DLT window**⁹.

Thall et al. (*Clinical Trials* 2024, methodologists’ response to Project Optimus): conventional Phase I designs cannot reliably identify safe and effective doses¹⁰.

Le Tourneau et al., *JNCI* 2009 · Hong et al., *NEJM* 2020 · Li et al., *JTO* 2022 · Thall et al., *Clin Trials* 2024 · Rivière et al., *Ann Oncol* 2015

Q&A POCKET — frequentist alternatives. If a methodologist asks “what about non-Bayesian combination designs?” the honest answer is that very few practical, deployed frequentist combination designs exist. The field has converged on Bayesian / model-assisted designs (POCRM, BOIN-COMB, BLRM, PIPE, surface-free) because combinations require borrowing strength across dose pairs and partial-order constraints — naturally handled by priors and posterior updating, awkwardly handled without them. The cleanest frequentist alternative in the literature is Conaway, Dunbar, & Peddada 2004 (*Biometrics* 60:661-669), order-restricted isotonic regression for combinations — theoretically clean but not widely deployed. 3+3 generalizations to combinations exist but inherit all the original problems plus combinatorial explosion. Sequential lead-in (find A then add B) is the dominant industry practice — that is what we are criticizing. Q&A POCKET — Rivière backup. The Rivière et al 2015 systematic review of 162 combination phase I trials gives the “almost never” claim a documented count: 88% used 3+3, all but one used a design developed for single-agent evaluation. If anyone challenges “almost never” as an unsupported universal, that is your number. 3+3 answered the question it was built for — the MTD of a cytotoxic under a monotonic toxicity assumption. For RAS combinations none of those assumptions hold. The deeper problem is structural and operational. In practice almost no one searches the joint dose surface. The protocol fixes agent A’s dose and adds agent B, so two monotherapy designs run in series and the joint surface is never characterized. The Amgen example is the clean one, and notice it is not a competence problem. Same sponsor used a model-based Bayesian logistic regression for sotorasib monotherapy in CodeBreak 100, but for the sotorasib + pembrolizumab combination in CodeBreak 100/101 they fixed the IO dose, explored only sotorasib across cohorts, and never modeled the two-drug surface. And the combination’s defining toxicity proved the point twice over: it was severe hepatotoxicity you could not predict from the single agent, and 88% of the Grade 3-4 events landed outside the 21-day DLT window — so a Cycle 1 design would have missed most of it. This single example demonstrates three of the four assumptions failing at once: joint surface unsearched, toxicity non-additive, late-onset toxicity outside Cycle 1. Even sophisticated sponsors collapse the combination to a one-drug search. Thall, Garrett-Mayer, Wages, Halabi, and Cheung made the formal version of this argument in their 2024 *Clinical Trials* paper, the SCT response to Project Optimus: conventional Phase I designs cannot reliably identify safe and effective doses. Next slide covers what replaces 3+3 in the toolkit.

Project Optimus: the regulatory ground has shifted

- FDA final guidance, 2024¹¹: *Optimizing the Dosage of Human Prescription Drugs... for the Treatment of Oncologic Diseases*
- **Maximum Tolerated Dose** → **a dose optimized on benefit-risk (the OBD)**.
- The expectation now is to **study more than one dose and justify the choice on efficacy and safety**, with **randomized dosage comparison** as the FDA's recommended approach.

This is no longer a methodological preference. It's the regulatory baseline.

FDA Guidance for Industry, 2024 (Project Optimus)

What you may not have caught is that the FDA has formalized this. The 2024 Project Optimus guidance moves oncology dose-finding from “maximum tolerated dose” to a dose optimized on benefit-risk — what the field calls the OBD. The guidance language is “optimized dosage,” and the field's OBD shorthand maps to it directly. If you submit a Phase I protocol today that only characterizes the MTD, your reviewers are going to push back. This is no longer a methodological preference held by biostatisticians; it's the regulatory baseline. Now, on what the guidance actually expects: the FDA's stated default is to take two doses into Phase II — the MTD and a dose below it — and use a randomized comparison to determine which is better on benefit-risk. For a single agent that works. It does not scale to combinations, because you cannot randomize across a two-dimensional joint dose grid. That is exactly where model-based joint designs stop being a preference and start being the only practical way to meet the benefit-risk standard, which is where the rest of this section goes.

Why MTD fails for targeted agents and immunotherapies

- **Pembrolizumab**: flat exposure-response 2→10 mg/kg; MTD never reached, efficacy plateaued at lowest dose tested¹²
- **CAR-T**: non-monotonic; high cell doses drive severe CRS/ICANS without proportional benefit
- **A RAS combination spans both**. Backbone on the plateau, IO/bispecific partner brings the umbrella. Neither rewards titrating to toxicity.

The reason MTD fails for these agents is mechanistic, and it has two flavors. I'll illustrate with pembrolizumab and CAR-T — non-RAS examples, but the cleanest in the literature, and the same dose-response shapes show up in the IO and bispecific partners daraxonrasib is going to be combined with. The first flavor you all know from pembrolizumab — the efficacy curve plateaus very early. In the original pembrolizumab studies, response rates across 2 mg/kg, 10 mg/kg every three weeks, and 10 mg/kg every two weeks were statistically indistinguishable. The exposure-response curve was flat. The MTD was never reached because efficacy never improved with dose. The 200 mg flat dose used today was selected by population PK modeling

to reproduce the exposures seen at 2 mg/kg in a typical patient — a weight-based-to-flat conversion of the lowest active dose, not a search for the optimum. The second flavor is the umbrella. CAR-T is the canonical case. Very high cell doses unambiguously drive severe CRS and ICANS without proportional benefit. The downslope itself is product-specific — some show a true decline, others a plateau or threshold — but what is universal across CAR-T and bispecifics is the absence of a monotonic increase, and that alone is enough to break titrate-to-toxicity. These are not RAS drugs, but they are the cleanest pictures of the two shapes, and they are the shapes your RAS combinations will actually have. Daraxonrasib is the plateau — its monotherapy program never cleanly reached an MTD, had no DLTs, and 300 mg was selected on tolerability, which is exactly why 300 versus 200 is still a live question. The partners you'll add bring the umbrella. A RAS combination lives across both panels, and that is the whole reason MTD logic cannot find its optimal dose. You need a design that watches both dimensions.

The cost of MTD-only thinking

Sotorasib was approved at the highest dose tested. A 4× lower dose — never compared during the trial — was later shown to give similar PFS with less toxicity.

- **Sotorasib** (KRAS G12C, first-in-class, approved 2021)⁸:
 - 960 mg selected as the **highest dose tested**, no DLTs — a more-is-better default, not an MTD
 - FDA required a **240 mg vs 960 mg** comparison. **PFS was similar** at one-quarter the exposure¹³
- **MTD optimal dose** for targeted agents and immunotherapies
- **More than 40% of patients** in small-molecule registrational trials require dose reductions or interruptions, a marker of overly aggressive dosing¹⁴

Same drug class as daraxonrasib.

Hong et al., *NEJM* 2020 (CodeBreaK 100) · Hochmair et al., *Eur J Cancer* 2024 · FDA–AACR, *Clin Cancer Res* 2025

The empirical evidence that this is not just theoretical. The FDA-AACR Strategies for Optimizing Dosages series, published in Clinical Cancer Research in 2025, reports that more than 40% of patients in small-molecule registrational trials require dose reductions or interruptions due to overly aggressive dosing — and links that pattern to potentially accelerated emergence of drug resistance, which quietly foreshadows the back half of this talk. The case you all know best is sotorasib, the first-in-class KRAS G12C inhibitor, approved in 2021. The Phase I program never reached an MTD because there were no DLTs; 960 mg was simply the highest dose tested, a more-is-better default rather than an MTD-driven choice. The FDA later required a

post-marketing comparison against 240 mg, a fourfold lower dose. PFS at 240 mg was similar to PFS at 960 mg — the clean evidence that the approved dose was four times higher than needed on the primary endpoint. Note: I am deliberately not folding in the October 2023 ODAC vote on CodeBreaK 200 here. That 10-2 vote was about whether the confirmatory PFS read-out could be reliably interpreted given trial-conduct issues — informative censoring, imbalanced early dropout, investigator-assessed imaging — not about the dose. Conflating it with the dose story is the move a regulatory-savvy attendee will flag. The dose story alone is airtight; keep it that way. For daraxonrasib — same drug class — getting this wrong in Phase I means every combination on top of it inherits an unnecessarily toxic starting point.

Combination dose selection in RAS is harder

Single agents already broke MTD logic. Combinations make it worse on every axis.

- You are no longer choosing one dose. You are choosing a **joint pair on a surface the single-agent data cannot reconstruct**⁸
- Effects interact on both efficacy and toxicity — the optimum can sit **below either single-agent dose**
- And for RAS, the **toxicity headroom is already spent** at the monotherapy dose²

Fix what you know. Search what you don't. Reduce to one drug only when the fixed component's dose is externally established — spend the search on the drug whose combination-optimal dose is genuinely uncertain.

Hong et al., *NEJM* 2020 · Wolpin et al., *NEJM* 2026 (RMC-6236-001)

This is the bridge from the empirical cost — sotorasib at the MTD-only dose, daraxonrasib's already-spent headroom at 300 mg — into the methodological answer that follows. I am deliberately not re-establishing single-agent MTD failure here; the last three slides have already done that work. The job of this slide is to name what changes when you go from one drug to two. Three things. First, you are no longer choosing one dose, you are choosing a joint pair, and the single-agent margins do not reconstruct the surface — sotorasib's 960-versus-240 result tells you the surface looks different from what the single-dose escalation suggested. Second, effects interact on both efficacy and toxicity — the synergy point from assumption 4 — so the joint optimum can sit below either monotherapy dose. Third, and this is the RAS-specific part: the monotherapy toxicity is already substantial, 34% G 3 at 300 mg, 48% dose modification, so you start the combination with less headroom to spend. The first combination cohort already moved daraxonrasib down to 200 mg with GnP, but we do not know whether 200 mg is right or whether 150 mg or some other dose is right given other partners. That is a dose-finding question that requires designs built for it. Which leads into the next slide.

Why Bayesian adaptive designs, and a primer on BOIN

Why Bayesian / model-assisted:

- Well suited to **small, sequential Phase I decision-making**
- Decision-theoretic framework supports **utility-based dosing**

BOIN family¹⁵:

- Simple as 3+3 (printed decision table)
- Statistically matches model-based designs
- **Targets a specific DLT rate** (e.g., 25%)
- Extends to **BOIN-COMB**¹⁶, **TITE-BOIN**¹⁷, **BOIN12**¹⁸
- **POCRM**¹⁹ and **BLRM**²⁰ are common combination alternatives

RAS adoption: sotorasib BLRM⁸, adagrasib mTPI (KRYSTAL-1)²¹.

Liu & Yuan 2015 (BOIN) · Lin & Yin 2017 (BOIN-COMB) · Yuan 2018 (TITE-BOIN) · Lin 2020 (BOIN12) · Wages 2011 (POCRM) · Neuenschwander 2008 (BLRM)

Brief framing of the toolkit that replaces 3+3. Two reasons Bayesian/model-assisted designs fit modern Phase I. First, they handle small sequential sample sizes by design — cohort-by-cohort posterior updates are the natural framework for Phase I decision-making. Second, they support utility-based dosing, which is how the OBD paradigm becomes operational: tell the trial up front how to trade off response against toxicity. BOIN, from Liu and Yuan in 2015, is the workhorse of the family. Operationally as simple as 3+3 — printed decision table at the bedside, PI reads off escalate/stay/de-escalate. Statistically matches the best fully model-based designs. And the family extends naturally: BOIN-COMB for drug combinations, TITE-BOIN for late-onset toxicity, BOIN12 for joint toxicity-efficacy. POCRM (Wages and Conaway) and BLRM are the common alternatives for combinations specifically. The bottom note is the credibility line: the leading RAS Phase I programs are already running on Bayesian designs. Sotorasib used a Bayesian Logistic Regression Model (Hong et al. NEJM 2020). Adagrasib used accelerated titration into mTPI in KRYSTAL-1 (Ou et al. JCO 2022 — citation TODO, add when verified). The methodological shift Project Optimus formalized is already industry standard for RAS monotherapy — the question is whether combinations follow.

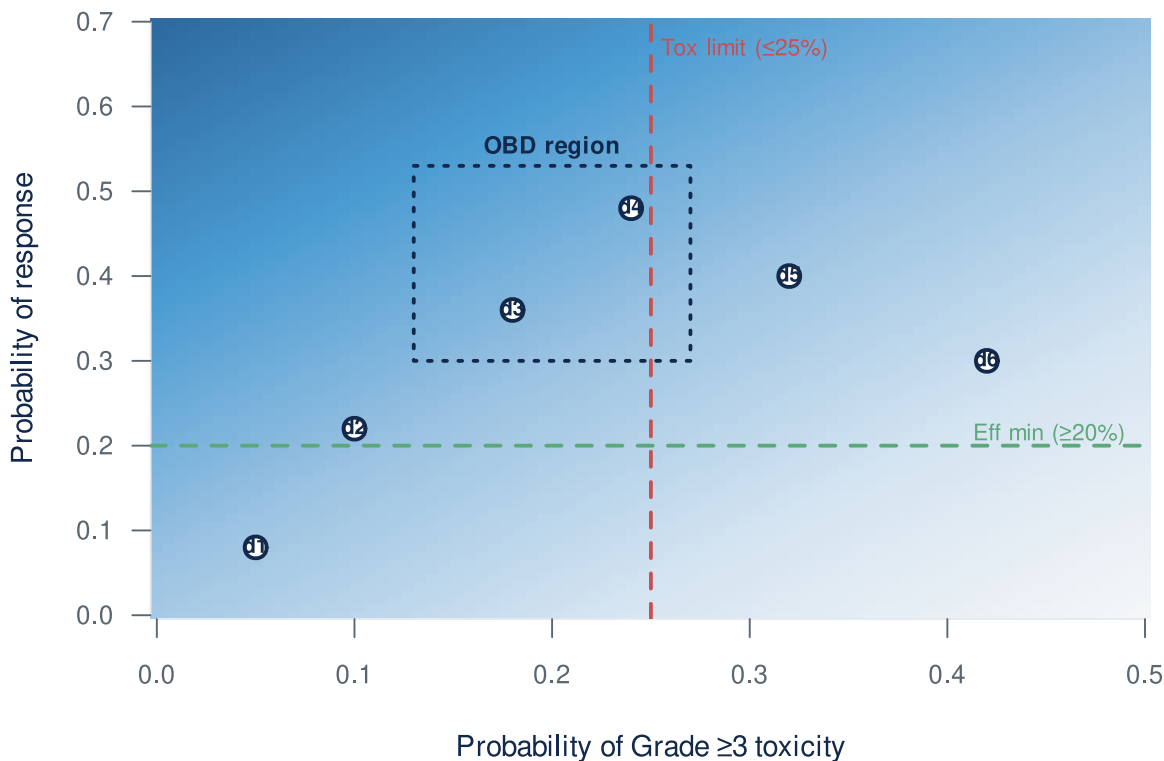
The new paradigm: OBD instead of MTD

Phase I no longer ends with a single MTD; it ends with an OBD.

U-BOIN²²:

- Up front: *“a response is worth this; a Grade 3 toxicity costs this”*
- **Stage 1:** BOIN rules establish the safe envelope (toxicity)
- **Stage 2:** utility-guided allocation finds the OBD *within* that envelope

Delivers the OBD Project Optimus calls for¹¹ — and reaches it where randomized dose comparison does not scale to combinations.



This is the new paradigm in one slide. Phase I trials no longer end with a single MTD that you carry into Phase II hoping for the best. They end with an OBD — an optimal biological dose — that has been characterized for both safety and efficacy during Phase I. U-BOIN is the cleanest example of what this looks like operationally. Before the trial starts, your team sits down and answers a question: how do you trade off response against severe toxicity? Response with no toxicity is the best outcome; severe toxicity without response is the worst; the intermediate combinations get scored in between. Those scores become the navigation system. Stage one uses BOIN's toxicity rules to establish the safe envelope. Stage two — the new part — allocates patients within that envelope to find the dose that maximizes the benefit-risk utility you specified. What comes out of Phase I is not the MTD. It's the OBD — a dose with a characterized benefit-risk profile. That's the dose you carry into Phase II combinations. That's what Project Optimus is asking sponsors to deliver. And in practice, this paradigm is increasingly the expected approach in modern oncology Phase I/II design.

PANGAEA: U-BOIN in basal-like PDAC

- **Trial:** LCCC2220-DCT (Somasundaram, PI), basal-like PDAC by **PurIST**²³

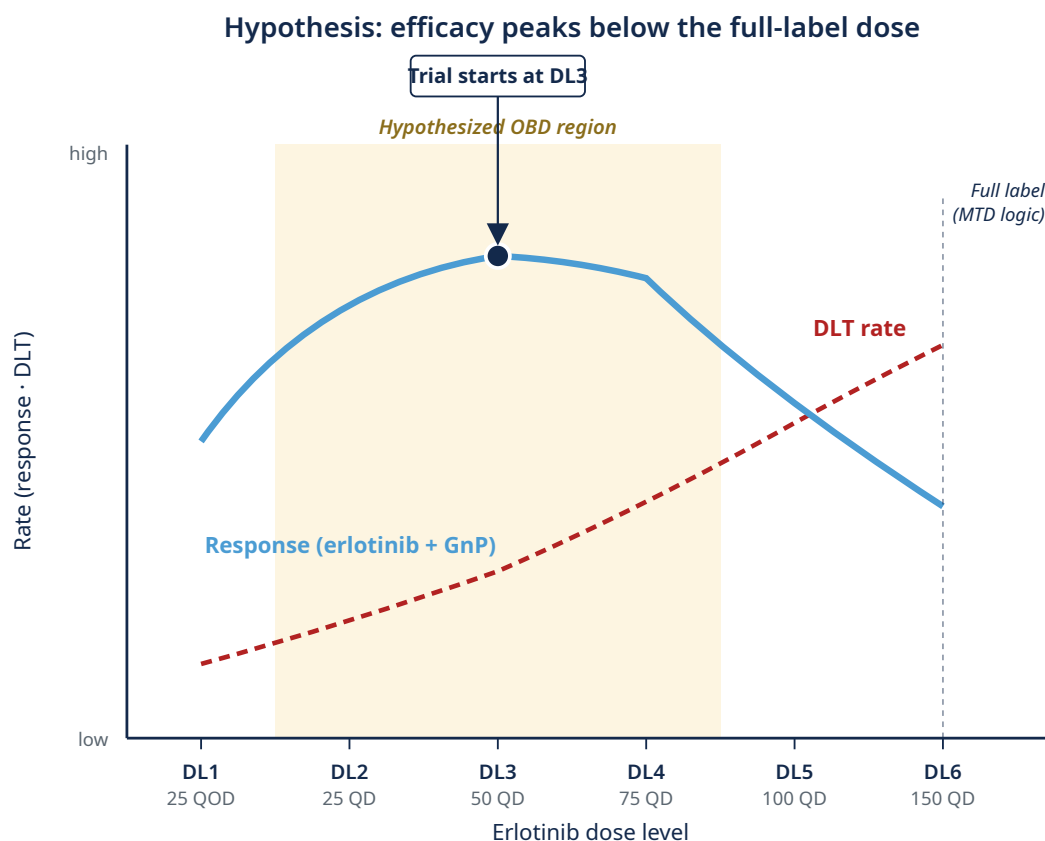
- **Combination:** erlotinib + biweekly gemcitabine/nab-paclitaxel
- **Hypothesis:** lower-dose erlotinib may be more efficacious (OBD-below-MTD)²⁴
- **Six dose levels:** 25 mg QOD → 25 / 50 / 75 / 100 / 150 mg QD; **start at DL3**

Design: *Stage 1 BOIN*¹⁵ (target DLT 25%, cohorts of 4); *Stage 2 BOIN12-like utility within U-BOIN*²² (tox 25%, eff 20%).

Up to 52 patients; ~28 at OBD (Scenario 1).

Justified dimensionality reduction: GnP fixed because the FDA-approved standard-of-care dose is established (MPACT); erlotinib searched because its combination OBD-below-MTD is the genuinely open question.

Rashid et al., *Clin Cancer Res* 2020 (PurIST) · Cohen et al., *Cancer Chemo Pharm* 2016 · Liu & Yuan 2015 · Zhou, Lee & Yuan 2019 (U-BOIN)



Here's the design in a real PDAC trial at UNC. PANGEA, led by Ashwin Somasundaram. The population is biomarker-defined — basal-like PDAC by PurIST, a subtype that has historically

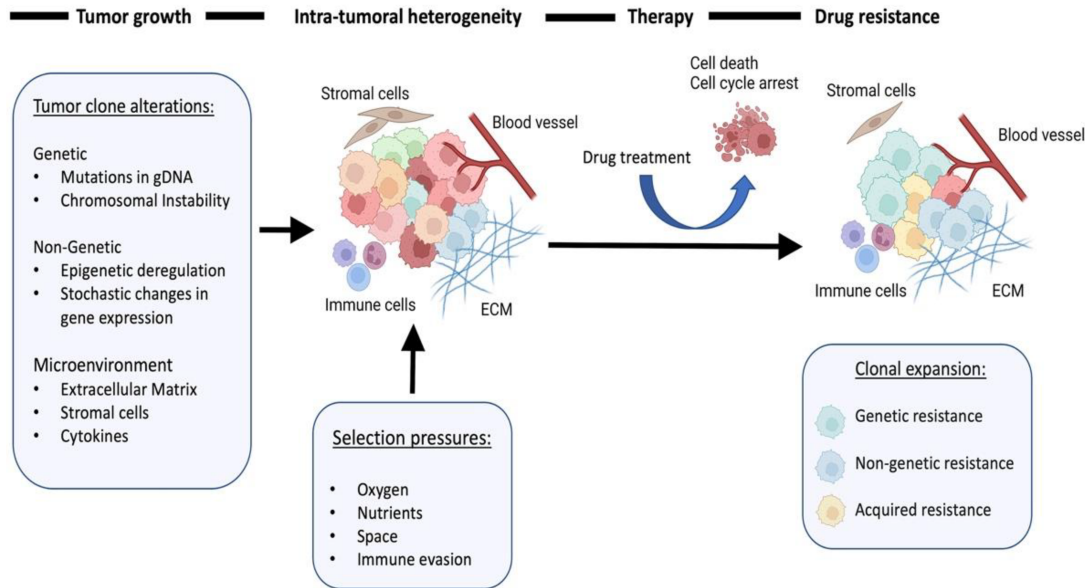
responded poorly to gemcitabine/nab-paclitaxel. The intervention is adding low-dose erlotinib to that backbone. And here's where the talk's argument lands on a real protocol: the central biological hypothesis of PANGEA is that lower doses of erlotinib may be more efficacious than the labeled dose, when given in combination with chemotherapy. The full-label dose of erlotinib is 150 mg daily. PANGEA tests six dose levels going down to 25 mg every other day. This is the OBD-below-MTD case made concrete — exactly the kind of question 3+3 cannot answer, and exactly the kind of question U-BOIN was built for. Stage 1 uses BOIN-style toxicity rules to establish the safe envelope, with a target DLT rate of 25%. Once we have enough data at any dose, we transition to Stage 2, which uses a utility function — elicited from our clinical team — to find the dose that maximizes the benefit-risk tradeoff. Doses have to clear both toxicity admissibility, no more than 25% DLT, and efficacy admissibility, at least 20% response rate. The trial ends when twenty patients have accumulated at any one dose, or when fifty-two patients have been treated total. Under the most plausible scenario from our simulations, we expect about thirty-eight patients in the U-BOIN phase, with about twenty-eight ultimately treated at the OBD. This is what an OBD-paradigm Phase I trial looks like in basal-like PDAC.

Resistance is the next problem

Each patient is a moving target.

Resistance is a temporal process

Classical trial design assumes patients are draws from a *fixed* distribution. Under RAS inhibition, **each patient is a moving target**, and a trial that doesn't *see* resistance arise cannot act on it.



Schematic adapted from ARPA-H ADAPT program overview materials.

Holding the thread from Act II — we still have a calibration burden waiting for us. But before we get there I want to layer on the second design problem these combinations create. Classical trials treat patients as if they're samples from a fixed distribution — you enroll them, you randomize them, you administer the regimen, you measure the endpoint at a fixed time. That logic works when the underlying biology is static. It doesn't work for RAS inhibitors, because under sustained RAS pressure the tumor evolves. Each patient is a moving target. Resistance signals start appearing weeks into treatment — sometimes earlier — and a trial that's only looking at fixed endpoints months down the line cannot see them in time to do anything about them. The design has to become temporal too.

Different resistance mechanisms require different detection strategies

Genomic → catchable by ctDNA · **Non-genomic / plasticity** → requires serial biopsies, transcriptomics, imaging · A resistance-aware trial has to *do both*.

Genomic resistance

- **Clinical PDAC** (Aguirre, *Cancer Discovery* 2024): PIK3CA, KRAS amplification, MYC/MET/EGFR/CDK6 amplifications²⁵
- **Preclinical RAS(ON)** (Wasko, *Nature* 2024): Myc amplification, PI3K family gains under RMC-7977²⁶
- **G12C clinical adagrasib** (Awad, *NEJM* 2021): NRAS, BRAF, MAP2K1; ALK/RET/BRAF/RAF1/FGFR3 fusions²⁷

Non-genomic: cell-state plasticity

- PDAC cells morph into RAS-independent states²⁵
- Dynamic kinome reprogramming (Stalnecker, UNC, AACR RAS 2026)²⁸

Aguirre et al., *Cancer Discov* 2024 · Wasko et al., *Nature* 2024 · Awad et al., *NEJM* 2021 · Xu et al., *Cancer Discov* 2025

The resistance landscape splits into two classes, and the audience here will see both of these in their clinics. But the design implication is what I want to lead with: different resistance mechanisms require different detection strategies built into the trial. Genomic resistance — NRAS mutations, MAP2K1 mutations, on-target mutations — is catchable by ctDNA. That's relatively cheap, continuous, easy to add to a trial. Non-genomic resistance — cell-state plasticity, kinome reprogramming, metabolic rewiring — needs serial biopsies, transcriptomic monitoring, sometimes imaging. That's expensive, episodic, hard. A resistance-aware trial has to design for both, because both are happening in your patients simultaneously. Andy Aguirre's group at Dana-Farber has published extensively on the genomic side. Closer to home, Clint Stalnecker here at UNC has shown the non-genomic side at the kinome and metabolic level. The biology is on the slide as evidence; the design implication is the headline.

Why classical trial designs miss resistance

- **Classical design was built for static biology** — fixed regimen, fixed endpoint.
- **In practice, resistance is studied retrospectively** — emerging signatures are characterized only after progression and trial closure^{25,27}
- The temporal surface (when resistance emerges, in which patients) is never characterized.

Same structural failure as combination dose-finding: static designs for a dynamic problem. A resistance-aware trial has to learn during enrollment.

Awad et al., *NEJM* 2021 · Aguirre et al., *Cancer Discov* 2024 (both post-progression analyses)

The resistance-side parallel to the combination dose-finding critique two sections ago. There, the problem was that 3+3 searches one dose at a time and never characterizes the joint dose surface. Here, the problem is that classical designs measure outcome at fixed endpoints and never characterize the temporal surface — when resistance emerges, in which patients, by which mechanism. The retrospective bullet is itself the evidence: the Awad and Aguirre papers I just cited on the previous slide are post-progression analyses, ctDNA and biopsies collected after patients had already failed therapy. The field's best resistance knowledge was generated exactly the way I'm criticizing — after the fact, outside any prospective design built to catch it. Early signals in molecular subsets dilute into the overall Kaplan-Meier. Heterogeneous response trajectories collapse into summary statistics. None of that is useful for designing the next trial in real time. The same structural failure as the dose problem: static designs for a dynamic problem. Just as the dose problem needed Bayesian designs that handle the joint

surface, the resistance problem needs designs that learn during enrollment. Next slide shows what those look like.

ADAPT / EVOLVE: what a resistance-aware trial looks like

- ARPA-H **ADAPT** program → **EVOLVE** trial in metastatic breast cancer (UNC-led; PI: Carey, Rashid co-PI)
- **Different disease, same design problem:** longitudinal resistance detection + biomarker-guided adaptation. The four axes below are disease-agnostic; a RAS-backbone PDAC combination would need all four for the same reasons.

Axis	Operationalized as
Adaptive stopping	Bayesian futility/efficacy boundaries updated at each interim
Real-time biomarker discovery	Pre-specified candidate signatures + agnostic discovery window
Longitudinal monitoring	Serial ctDNA, imaging, biopsies on every patient — <i>the “do both” requirement</i>
In-trial evaluation	Biomarker-guided cohorts within the same protocol — <i>characterize the temporal surface and act on it</i>

This is happening, federally funded, enrolling.

This is what a resistance-aware trial actually looks like operationally. ARPA-H — the new Advanced Research Projects Agency for Health — has funded a program called ADAPT specifically to build trials of this kind. Our EVOLVE trial in metastatic breast cancer is one of the first, and the design template generalizes directly to the kind of RAS-backbone combination trial you all are going to be running for PDAC in the next few years. I am showing a breast cancer trial to GI SPORE leaders because the four axes below are disease-agnostic — they’re requirements for any resistance-aware trial, not properties of breast biology. A RAS-backbone PDAC combination would need all four for the same reasons. Four axes. Adaptive stopping rules — Bayesian futility and efficacy boundaries that update at every interim look. Real-time biomarker discovery — pre-specified candidate signatures plus an agnostic discovery window so you’re not locked into your initial hypothesis. Longitudinal monitoring — serial ctDNA, imaging, biopsies on every patient. This is the “do both” requirement from the mechanisms slide made concrete: ctDNA for the genomic side, biopsies and transcriptomics for the non-genomic plasticity side, on every patient throughout the trial. And in-trial evaluation — biomarker-guided cohorts opened within the same protocol, so when you find a resistance signature you can test the strategy that addresses it during the trial, not as a follow-on study three years later. This is the answer to “characterize the temporal surface and act on it during enrollment”

that I named two slides ago. CRITICAL FRAMING DISCIPLINE: the box says “enrolling,” not “EVOLVE shows” or “EVOLVE works.” This is the resistance-side honesty I committed to in the opening — methods exist and are deployed, not yet completed. If a trialist asks for the readout, the answer is the design exists and is accruing; the readout is what we will report when interim looks are run. Do not let the live delivery drift toward an implied result.

Putting these into practice

Why don't we see more of these trials?

Why don't we see more of these trials?

Funding, accrual, and biopsy logistics are real.

But the barrier that quietly kills good designs — and the one we can actually remove — is calibration.

- Modern designs have many parameters: DLT/efficacy targets, utility weights, stopping rules
- Operating characteristics (Type I, power, N, duration) depend on every parameter, **in interaction**
- Each candidate: thousands of simulated trials $\times$ scenarios
- Design space: **8–13 dimensions** for the trials we've calibrated, more for combined dose + resistance. Grid search is infeasible.
- Field has adopted the machinery: Bayesian designs in **48% $\rightarrow$ 75% of Phase I protocols (2021–2024)**, but dose-optimization plans in only **30%**²⁹

In our group's experience, several months per design. For combined dose + resistance designs (20+ parameters), hand-tuning is infeasible.

FDA–AACR Project Optimus impact review, *JCO Onc Adv* 2025 (367 Phase I protocols)

This is the slide that opens the BATON section, and I want to land it carefully. Open with the concession explicitly: funding, accrual, biopsy logistics — these are real, and this room has lived them. I am not standing up here telling you the barrier to better trials is statistics, because that would be a biostatistician naming his own contribution as the field's central obstacle, and you would be right to push back. What I am claiming is narrower and more specific: among the barriers, calibration is the one that quietly kills good designs and the one we can actually remove. You have all watched PIs default back to 3+3 even when they knew a better design existed — the calibration cost was the hidden reason, and that is the gap BATON closes. Now the body. The methods I just walked through — modern dose-finding for combinations, longitudinal biomarker-aware adaptive designs — exist. The reason every clinical trial isn't using them is calibration cost. Every modern design has knobs: target DLT and efficacy rates,

utility weights, cohort sizes, stopping rules, sample-size caps. Operating characteristics depend on every knob, in interaction. To know whether a design meets Type I, power, expected N, expected duration, OBD-selection accuracy, and biomarker-promotion error control, you simulate thousands of trial replicates per candidate. And you do this across plausible scenarios. The design space is eight to thirteen dimensions for the trials we’ve calibrated, and twenty-plus when you combine dose-finding with resistance tracking. A grid search is hopeless — both computationally infeasible and statistically wasteful, because most of the parameter space gives bad designs. In my group, calibrating a single modern combination Phase I/II design by hand takes a senior statistician several months of focused work. For trials that handle dose-finding AND resistance tracking simultaneously, you’re at twenty-plus parameters with even more constraints. Hand-tuning doesn’t just get slower; it gets infeasible. That’s the gap BATON closes.

BATON: the calibration tool

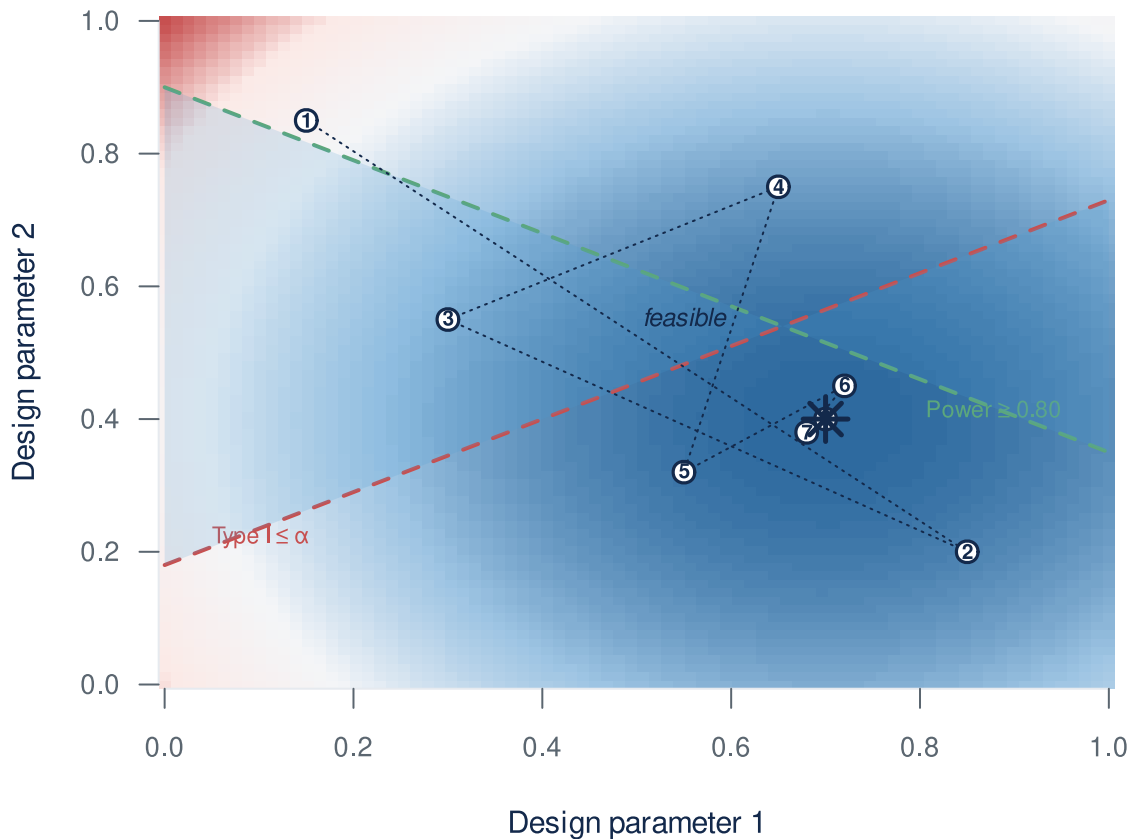
Our answer to the calibration question:

BATON = Bayesian Adaptive Trial Optimization³⁰

“BATON automatically searches the design parameter space to find trial configurations satisfying all operating-characteristic constraints — minimizing expected N, max N, or a trade-off, depending on the loss you specify.”

BATON tunes the design; it does not replace it.

You bring the clinical design; BATON finds the parameter settings that meet your targets. Open-source R package: github.com/naimurashid/BATON



Red = high expected N (bad); blue = low. Dashed lines = Type I and Power constraints. Numbered points = BATON's sequential evaluations converging to the constrained optimum ().

Young et al., 2026 (BATON manuscript) · github.com/naimurashid/BATON

BATON does one thing, and I'm going to tell you what it is in a single sentence. BATON automatically searches the design parameter space to find trial configurations that satisfy all the operating-characteristic constraints — Type I, power, sample size, duration, all of them — under whichever loss you specify, whether that's minimum expected N, minimum max N, or a trade-off between them. That's it. One important distinction before I show you results: BATON does not design the trial. You and your clinical team bring the clinical design — the adaptive structure, the stopping rule logic, the endpoints, the biomarker strategy. BATON takes that design and finds the parameter settings that meet your operating-characteristic targets. Under the hood there's constrained Bayesian optimization with Gaussian process surrogates, but for the clinical audience what matters is the input and the output. You give BATON a design template, a set of operating-characteristic targets, and a scenario grid. BATON returns a calibrated design that meets your targets. Days instead of months. FRAMING DISCIPLINE — and this matters because the next slide is PANGAEA: in the BATON manuscript, the worked

example is EVOLVE, not PANGEA. BATON was developed and validated during EVOLVE activation, across breast cancer subtrials at 8 to 13 parameters. PANGEA is an application of the same tool, not what the paper reports. If anyone asks “is this published,” the clean answer is: the method and its validation are under review at JASA Applications and Case Studies; PANGEA is one application of the same software. Don’t let the live delivery imply the PANGEA calibration is what the paper validates. Q&A POCKETS, in order of likelihood. (1) “Is BATON really optimal?” Manuscript benchmarked it against Simon two-stage designs where the optimum is known analytically; BATON converges to within 1% of Simon-optimal expected N, hitting the optimum exactly in 8 of 10 seeded runs and recovering the optimal design across most of 12 scenarios. So the answer is: it recovers the analytically optimal design on problems where we can check. The “near-optimum on plateaus” caveat is a footnote, not a retreat. (2) “Can we use it?” Yes — open-source R package at github.com/naimurashid/BATON, with the companion `evolveTrial` package. (3) “What’s the gain over grid or random search?” The paper has a head-to-head: BATON found feasible designs in 150–200 evaluations where grid and random search often found none at all. That’s the empirical version of the “calibration is the barrier” argument made on the previous slide.

PANGEA Phase II: calibrated by BATON

- After Phase I identifies the OBD, PANGEA continues into a **Phase II Bayesian adaptive randomized trial**: GnP + erlotinib (at OBD) vs GnP alone
- Primary endpoint: PFS. **Design assumes** reference median 5.5 mo³¹; powered to detect 9.2 mo under H (HR 0.58)
- **BATON-calibrated** four parameters (N, efficacy threshold, futility threshold, futility margin)³⁰
- **Constraints met**: Type I 0.10, Power 0.80

Achieved operating characteristics:

Metric	Value
Power	82.2%
Type I error	7.0%
Max N	76
Expected N under H	53.0
Expected N under H	58.1
Pr(early futility stop H)	77.9%
Pr(early efficacy stop H)	81.4%
Expected duration	27.3 mo

Von Hoff et al., *NEJM* 2013 (MPACT) · Young et al., 2026 (BATON manuscript)

Once U-BOIN selects the OBD, PANGAEA transitions into a Phase II Bayesian adaptive randomized trial — basal-like PDAC patients randomized one-to-one to GnP plus erlotinib at the OBD versus GnP alone, with PFS as the primary endpoint. TENSE DISCIPLINE — this is critical and the slide structure protects it but the live delivery needs to as well. The 5.5-month reference, the 9.2-month alternative, and the HR 0.58 are DESIGN ASSUMPTIONS, not observed results. The 5.5 is the GnP doublet benchmark from MPACT, Von Hoff 2013. The trial is powered to detect a hazard ratio of 0.58, not showing one. The operating characteristics in the right column ARE achieved, because they are the outputs of BATON’s calibration — Type I at 7.0%, power at 82.2%, those are properties of the design that BATON found, computed over 10,000 simulated trials per scenario. So: design assumptions on the left, calibrated achievements on the right. Never let “BATON found a design that achieves these OCs” drift to “the trial showed this.” The design uses a Bayesian piecewise exponential survival model with interim analyses for early stopping on futility or efficacy. Four design parameters had to be set jointly to meet operating characteristics — the maximum sample size, the efficacy stopping threshold, the futility stopping threshold, and the futility hazard-ratio margin. CONFIRM AGAINST MANUSCRIPT before delivery which exact four BATON searched versus which were fixed by clinical/modeling choice, since a methodologist may ask. BATON found a configuration that meets Type I error at most 0.10 and power at least 0.80, simultaneously. The achieved characteristics are on the slide. Three numbers I want to draw your attention to. First, the power is eighty-two percent at a Type I error of seven percent — solidly within the targets. Second — and this is the clinical payoff — the trial stops early for futility seventy-eight percent of the time when the drug doesn’t work. Patients are not being randomized to a losing arm waiting for the readout. Third, the trial stops early for efficacy eighty-one percent of the time when the drug does work. That’s how quickly modern adaptive designs can deliver an answer when there’s signal. Expected trial duration is twenty-seven months, with expected enrollment of fifty-three patients under the null and fifty-eight under the alternative — well below the calibrated max of seventy-six. (Note: the protocol narrative carries 82 as the accrual target with buffer; 76 is the calibrated design maximum. If anyone asks, “76 is the calibrated maximum, 82 is the accrual goal.” Action item for me separately: reconcile the protocol so a reader does not catch the discrepancy.) This entire calibration was done by BATON. Hand-tuning a design with these four interacting parameters against two simultaneous operating-characteristic constraints, across thousands of simulated trials, is the work I was describing earlier as several months of senior statistician time. BATON delivered this configuration in hours.

Where this is going

- **Today:** BATON has calibrated **EVOLVE** (single-arm, between-arm, seamless two-stage subtrials) and **PANGAEA Phase II**³⁰.
- **Next:** Extend BATON to (i) **Phase I Bayesian adaptive designs** (U-BOIN, BOIN12) for OBD selection; (ii) **joint-surface combination designs** (POCRM¹⁹, BOIN-COMB¹⁶, BLRM²⁰) for novel-novel RAS doublets (e.g., daraxonrasib + elironrasib) where neither

dose is externally established; (iii) scenarios where **trial assumptions are uncertain** (CARBO).

- The barrier pushing PIs back to 3+3, *gone*.

The next generation of GI oncology trials should not be limited by what a statistician can hand-tune.

Young et al., 2026 (BATON) · Wages 2011 (POCRM) · Lin & Yin 2017 (BOIN-COMB) · Neuenschwander 2008 (BLRM)

Q&A POCKET — non-RAS examples of joint-surface combination dose-finding. If anyone asks “do any trials actually use joint-surface combination designs”: yes, two clean concrete answers. (1) POCRM: the Embolden trial (NCT03240211) at UVA — pembrolizumab + pralatrexate/decitabine combination Phase Ib in T cell lymphomas, published implementation by Mozgunov et al, Stat Med 2022. (2) BLRM: Novartis Oncology has used BLRM as its standard Phase I design for over a decade and now exclusively for Phase 1 — including combination trials via the dual-agent extension. So joint-surface designs are not theoretical — they have published RCT deployments and one of the largest oncology developers in the world uses BLRM by default. The RAS-specific gap is that novel-novel RAS doublets like elironrasib + daraxonrasib are in the clinic but have not yet adopted these methods. That gap is exactly where BATON’s joint-design extension lands. Where this is going, in two pieces. First, what’s done. BATON has calibrated both EVOLVE — the single-arm, between-arm, and seamless two-stage breast cancer subtrials reported in the BATON manuscript — and PANGAEA Phase II, the calibration you just saw. Two trials, two diseases, same tool. Second, the extensions in flight. The methodological move that matters most for this audience is bringing BATON’s constrained-Bayesian-optimization machinery to the Phase I family — U-BOIN and BOIN12 — for OBD selection. PANGAEA Phase I uses U-BOIN today, calibrated with the standard online software; the next step is doing that calibration under BATON so the multi-parameter Phase I design space stops being the bottleneck. That’s how the calibration cost that has historically pushed PIs back toward 3+3 disappears. The second extension is CARBO — Context-Aware Robust Bayesian Optimization — which calibrates against a range of plausible trial assumptions rather than a single point estimate. This matters when the assumption was X but the trial actually enrolled at half the rate with 60% of the assumed biomarker prevalence; CARBO makes the design robust to that uncertainty. Mention CARBO in Q&A territory if time permits. The bigger point: when a modern Phase I/II design can be calibrated in days instead of months, the design choice becomes about what’s right for the trial, not what’s feasible in the timeline. The next generation of GI oncology trials should not be limited by what a statistician can hand-tune. We have the methods to do better, and the infrastructure to deploy them.

Take-home

1. **The RAS era needs new kinds of trials.** Biomarker-guided, dose-optimized below MTD, resistance-aware.
2. **The methods exist.** Bayesian adaptive designs handle non-monotonic dosing, combinations, and biomarker-guided adaptation.
3. **BATON closes the deployment gap.** Days instead of months.

New RAS therapies need new kinds of trials. Methods to optimize combination doses are established. Designs that track resistance and adapt the trial in response are enrolling. What's missing is making them practical to calibrate and run.

Three things to leave with. First, the RAS era needs a new generation of trials: biomarker-guided combinations, dose-optimized below the MTD where non-monotonic dose-response makes the MTD the wrong target anyway, and resistance-aware because patients fail therapy and we need to see it happen during the trial, not after. Second, the methods to run those trials exist. Bayesian adaptive designs are the tool family — they handle non-monotonic dose-response through utility-based OBD-finding, they handle combinations through joint admissibility on toxicity and efficacy, and they handle in-trial biomarker adaptation through pre-specified rules with simulation-calibrated error control. Third, the bottleneck has been making these designs practical to deploy. BATON is our answer to that. It calibrates the parameter settings against your operating-characteristic targets in days instead of months, so design choices become about what's right for the trial rather than what's feasible in the timeline. The thesis I opened with: new RAS therapies need new kinds of trials. Methods exist. The bottleneck is making those designs practical to calibrate and run. Thank you.

Acknowledgments

Clinical collaborators

- Lisa Carey (UNC, EVOLVE PI)
- Channing Der (UNC, NEJM editorial³²)
- Jen Jen Yeh (UNC Surgery, STOP SPORE PI · NEJM editorial · PDAC collaborator)
- Ashwin Somasundaram (PANGAEA PI)

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Der & Yeh, *NEJM* 2026 (editorial)

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Questions

Thanks. Happy to take questions.